

Final coordinate-based model

A compact overview of the exact, integer-grid MILP for loading identical Euro pallets.

Modeling choice

Individual boxes are represented by pallet assignments, orientations, and direct coordinates. A complete enumeration of all positions b_{ipxyzo} is avoided. The model is exact relative to the chosen integer grid.

Component	Decision in the model
Geometry	Integer coordinates (x, y, z) and permitted orthogonal orientations
Non-overlap	Six possible separating directions for each box pair and pallet
Stability	Direct contact and at least 75% shared support area
Primary objective	Minimize the number of pallets used
Prepared extensions	Weight, product group, and retrieval priority as later modules

Core assumptions

- Rigid, axis-parallel cuboids; all boxes are known before optimization.
- Identical pallets with $L = 120$ cm, $W = 80$ cm, and usable height $H = 180$ cm.
- Integer coordinate grid, 1 cm by default; touching is allowed, volumetric overlap is not.
- Every box is loaded exactly once; permitted orientations may differ by box.
- At least 75% of each base area is supported by the pallet or by directly contacting boxes.
- The current solver uses one objective; multiple criteria are added later as modular extensions.

1. Sets, data, and variables

Index i denotes an individual box. Box types are used by the instance generator only to create identical box attributes; the exact formulation operates on individual boxes.

Symbol	Type	Meaning
$I = \{1, \dots, n\}$	set	Individual boxes
$P = \{1, \dots, P_{\max}\}$	set	Potential identical pallets
O_i	set	Permitted orientations of box i
l_{io}, w_{io}, h_{io}	parameter	Dimensions of i in orientation o
m_i, g_i, ρ_i	parameter	Weight, product group, and retrieval priority
u_p	binary	1 exactly when pallet p is used
a_{ip}	binary	1 exactly when box i is assigned to pallet p
r_{iop}	binary	1 exactly when orientation o is selected
x_{ip}, y_{ip}, z_{ip}	integer	Lower-left-rear corner of the box
$\lambda_{ijp}, \phi_{ijp}, \delta_{ijp}$	binary	i is left of / in front of / below j
O^x, O^y, O^{xy}	binary	Horizontal interval or footprint overlap
c_{ijp}	binary	Box i directly supports box j
g_{ip}	binary	Box i rests directly on the pallet floor

Orientation-dependent dimensions

$$l_{ip} = \sum_{o \in O_i} l_{io} r_{iop}, \quad \bar{w}_{ip} = \sum_{o \in O_i} w_{io} r_{iop}, \quad \bar{h}_{ip} = \sum_{o \in O_i} h_{io} r_{iop}$$

Interpretation

If $a_{ip} = 0$, all orientation variables and therefore the effective dimensions for this pallet are zero. If $a_{ip} = 1$, exactly one orientation is selected.

2. Assignment, orientation, and pallet boundaries

Every box is assigned to exactly one pallet:

$$\sum_{p \in P} a_{ip} = 1 \quad \forall i \in I$$

A pallet is activated as soon as a box is assigned to it:

$$a_{ip} \leq u_p \quad \forall i, p; \quad u_p \leq \sum_{i \in I} a_{ip} \quad \forall p$$

Exactly one orientation is selected if the box is assigned to the pallet:

$$\sum_{o \in O_i} r_{iop} = a_{ip} \quad \forall i, p$$

Pallet boundaries

$$x_{ip} + l_{ip} \leq L a_{ip}, \quad y_{ip} + \bar{w}_{ip} \leq W a_{ip}, \quad z_{ip} + \bar{h}_{ip} \leq H a_{ip}$$

For $a_{ip} = 1$, all box edges remain inside the pallet. For $a_{ip} = 0$, these constraints and non-negativity jointly enforce $x_{ip} = y_{ip} = z_{ip} = 0$.

Strengthening volume condition

$$\sum_{i \in I} V_i a_{ip} \leq LWH u_p \quad \forall p; \quad \sum_{p \in P} u_p \geq \lceil (\sum_i V_i) / (LWH) \rceil$$

Important

The volume condition does not replace the geometric constraints. It is only a fast necessary condition that improves the solver's lower bound.

Symmetry breaking

$$u_p \geq u_{p+1}; \quad \sum_i V_i a_{ip} \geq \sum_i V_i a_{i,p+1}$$

Used pallets appear without index gaps and are ordered by loaded volume. Fully identical boxes are additionally ordered by pallet index and a scalar position value.

3. Non-overlap and exact directional logic

For every unordered box pair $i < j$ and every pallet, at least one of the six spatial separation directions must hold.

$$\lambda_{ijp} + \lambda_{jip} + \phi_{ijp} + \phi_{jip} + \delta_{ijp} + \delta_{jip} \geq a_{ip} + a_{jp} - 1$$

Forward implications

$$\begin{aligned} x_{ip} + l_{ip} &\leq x_{jp} + L(1 - \lambda_{ijp}) \\ y_{ip} + w_{ip} &\leq y_{jp} + W(1 - \phi_{ijp}) \\ z_{ip} + h_{ip} &\leq z_{jp} + H(1 - \delta_{ijp}) \end{aligned}$$

Example: $\lambda_{ijp} = 1$ forces the right edge of i to end no later than the left edge of j . If the variable is zero, the corresponding inequality is deactivated by the tight Big-M value L , W , or H .

Reverse implications on the integer grid

$$x_{jp} \leq x_{ip} + l_{ip} - 1 + L(\lambda_{ijp} + 2 - a_{ip} - a_{jp})$$

If both boxes are on p and λ is zero, j must begin at least one grid unit before the end of i . Together with the forward implication, conditional on joint assignment, this yields:

$$\lambda_{ijp} = 1 \Leftrightarrow x_{ip} + l_{ip} \leq x_{jp}$$

Why the reverse implication?

Without it, λ , ϕ , and δ would only be freely selectable separation certificates. For overlap, accessibility, and product-placement logic, however, they must represent the actual relative positions. The -1 term is exact only for integer coordinates.

Horizontal overlap

$$O_{ijp}^x = a_{ip} \wedge a_{jp} \wedge (1 - \lambda_{ijp}) \wedge (1 - \lambda_{jip})$$

$$O_{ijp}^{xy} = O_{ijp}^x \wedge O_{ijp}^y$$

If two boxes merely touch at a side face, overlap in that direction is zero. Positive footprint overlap instead implies $O^{xy} = 1$.

4. Blocking and 75% support

Vertical blocking

$$v_{ijp} = \delta_{ijp} \wedge O_{\{i,j\}p}^{xy}$$

$$v_{ijp} \leq \delta_{ijp}, \quad v_{ijp} \leq O_{\{i,j\}p}^{xy}, \quad v_{ijp} \geq \delta_{ijp} + O_{\{i,j\}p}^{xy} - 1$$

Thus i lies completely below j and both occupy the same vertical retrieval corridor. Direct contact is not required; a gap or another box may lie between them.

Direct support relationship

$$c_{ijp} = 1 \Rightarrow z_{jp} = z_{ip} + \bar{h}_{ip}, \quad c_{ijp} \leq \delta_{ijp}, \quad c_{ijp} \leq O_{\{i,j\}p}^{xy}$$

The contact equality is activated by two Big-M inequalities. A separate variable g_{jp} detects exactly whether $z_{jp} = 0$, in which case the pallet itself provides support.

Overlap area of two footprints

$$q_{ijp}^x = \max\{0, \min(x_i + l_i, x_j + l_j) - \max(x_i, x_j)\}$$

$$q_{ijp}^y \text{ analogously; } A_{ijp} = q_{ijp}^x q_{ijp}^y$$

Minima, maxima, and positive parts are linearized with selector variables. The product is represented exactly through the binary expansion of q^x . The countable contact area $\hat{A}_{ijp}$ equals A under direct contact and zero otherwise.

Minimum support area

$$4 \sum_{i \neq j} \hat{A}_{ijp} + 4LW g_{jp} \geq 3B_{jp}$$

Here $B_{jp} = \sum_o l_{jo} w_{jo} r_{jop}$ is the base area under the selected orientation. For a box above the floor, the condition reduces to $\sum \hat{A} \geq 0.75 B$. Several boxes in direct contact may contribute jointly.

Scope

The 75% area rule is a static geometric surrogate. It does not model friction, deformation, tipping moments, acceleration, or maximum compressive loads.

5. Objective, extensions, and solution

Currently implemented objective

$$\min F_p = \min \sum_{p \in P} u_p$$

For identical pallets and complete loading, volume utilization $\eta = (\sum V_i) / (LWH \sum u_p)$ is determined directly by the number of pallets and is therefore reported rather than optimized separately.

Later soft extensions

Criterion	Linear measure	Interpretation
Compactness	$F_H = \sum_p T_p$	Total occupied pallet height
Heavy boxes low	$F_W = \sum m_i (z_{ip} + \bar{h}_{ip} / 2)$	Vertical weight moment
Food/chemicals	$F_C = \sum v_{ijp}$	Penalize food located below chemicals
Accessibility	$F_A = \sum w_{ij}^A v_{ijp}$	Penalize a later-needed box above an earlier box

Recommended multi-objective strategy

First minimize the pallet count exactly. Then fix its optimal value and optimize normalized secondary criteria by weighted sums or epsilon-constraints. This part is deliberately not yet activated in the current Python prototype.

Exact solution method

- Gurobi Branch-and-Bound / Branch-and-Cut on the integer coordinate model.
- The volume lower bound as a starting point and a heuristic solution as an upper bound.
- Symmetry breaking for pallets and fully identical boxes.
- Proof of optimality for small demonstration instances; report the time limit and MIP gap transparently.

Scaling and transition to heuristics

Small instance Exact MILP	Validation Optimal value and bounds	Large instance Extreme points / beam search
8-15 boxes initially	Compare heuristic with optimum	Dynamic candidates instead of a grid

No monolithic exact model will reliably solve roughly 1,000 heterogeneous boxes with all practical requirements. The MILP therefore serves as a precise problem definition, an exact benchmark for small instances, and a foundation for later geometric heuristics.

Final model in one line

Assignment + orientation + coordinates + pairwise separation + exact support area + minimization of the pallet count. Attributes for weight, product group, and priority remain in the instance format and can be added modularly.